Deviations Register — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
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1. Purpose

The Sprint Plan requires every departure from the SRS, SAD, Test Plan or Sprint Plan to be recorded here: what changed, why, and which requirement it affects. Each entry names the pull request or issue where the decision is visible in the repository history. Requirement and test-case IDs are the ones used in `Documents/SRS.pdf` and `Documents/Test_Plan.pdf`.

2. Summary

ID	Date	Baseline document	Affects	Kind	Status
D-01	22 Oct	SRS	FR-004, UT_014	Requirement interpretation	Accepted
D-02	22 Oct	SRS	FR-010, FR-023, NFR-010, UT_030	Design	Accepted
D-03	23 Oct	Sprint Plan	FR-019, FR-020	Scope moved earlier	Accepted
D-04	23–27 Oct	SRS (Organisational)	NFR-006	Process	Closed with exceptions; corrective action in Sprint 2
D-05	23–27 Oct	Sprint Plan	12 of 52 Sprint 1 points	Schedule	Resolved 27 Oct
D-06	23 Oct	Sprint Plan	Sprint 1 goal; FR-024, FR-027	Scope	Open; delivered by Sprint 2
D-07	19–27 Oct	Sprint Plan (Definition of Done)	All Sprint 1 test cases	Process	Open; Test Plan to be filled by 30 Oct

3. Deviations

D-01 — Course code is unique per semester, not globally

Baseline: FR-004 (SRS) says a duplicate course code is rejected.
Built: a course code is unique within a semester. The same code can be reused in a different semester. The database enforces this with the constraint `course_code_unique_per_semester`.
Why: the SRS and the Test Plan disagree. UT_014 expects the same code to be accepted in another semester, which matches how PES University reuses course codes every year. The team followed the test case.
Affects: FR-004, UT_014.
Recorded in: PR #71 (Komal), merged 22 Oct.

D-02 — Negative marking: toggle per examination, deduction per question

Baseline: FR-010 lists "negative marking (on/off and value)" as an examination policy value.
Built: the examination stores only the on/off switch (`Exam.negative_marking`). The deduction value is stored on each question (`Question.negative_marks`), next to its positive marks.
Why: FR-008 already gives every question its own marks. Keeping the deduction beside them lets a 1-mark and a 4-mark question carry different penalties in one paper. The evaluator applies a question's deduction only when the

examination's switch is on, so a single paper can still have negative marking turned off. The SRS's open decision on negative marking (flat deduction of the question's configured negative marks) is unchanged.
Affects: FR-010, FR-023, NFR-010. UT_030 still passes: the switch and a 0.5 deduction persist and read back as set.
Recorded in: PR #74 (Komal), merged 22 Oct.

D-03 — Answer saving delivered one sprint early

Baseline: the Sprint Plan places FR-019 (autosave every response) in Sprint 2.
Built: PR #73 (FR-020, automatic submission) also added the service that saves an answer and its endpoint POST `/api/attempts/<id>/answers`. Each choice is saved the moment it is made.
Why: FR-020 submits the answers held at expiry. Without saved answers, an attempt closed by the timer would always score zero.
Affects: FR-019 (Sprint 2 now adds `marked_for_review` and the 5-second guarantee on top of the existing endpoint), FR-020.
Recorded in: PR #73 (Kshitij), merged 23 Oct.

D-04 — Five pull requests merged without a second member's approval

Baseline: NFR-006 requires every merge to `main` to go through a pull request reviewed by another member. Its acceptance test: no direct commits to `main` after 19 Oct, and every merged PR has a reviewer other than its author.
What happened: there were no direct commits to `main`. Nine of the fourteen pull requests merged since 19 Oct carry an **Approve** review from another member. Five of Kshitij's pull requests were merged with the repository owner's admin override instead:

PR	Item	Review	Merged
#68	FR-022, FR-023 evaluation	Komal commented ("Looks good"), did not approve	23 Oct
#70	FR-015, FR-016 start attempt	Komal commented ("Cool!"), did not approve	23 Oct
#73	FR-020 automatic submission	Komal commented ("Okay"), did not approve	23 Oct
#79	Start exam button on the student exam page	None	27 Oct
#80	Shared stylesheet and role navigation	None	27 Oct

Why: #68, #70 and #73 were stacked (each built on the one before) and blocked the Sprint 1 video due 23 Oct. #79 and #80 were merged before the Sprint 1 review demo on 27 Oct. CI was green on all five, and the full suite (109 tests) passes on `main`.
Corrective action: from Sprint 2 there are no admin merges. Every pull request, including the repository owner's, waits for an **Approve** review from another member; a comment does not count.
Recorded in: issue #38, closed 27 Oct with the full audit.

D-05 — Sprint 1 finished four days late

Baseline: Sprint 1 runs 19–23 Oct with a 52-point commitment.
What happened: 40 points closed by 23 Oct. The remaining 12 points were coded within the sprint but merged on 27 Oct:

Item	Points	Closed by	Cause
FR-006 enrolled-exam visibility	3	PR #75	Review asked for 403 instead of 404 (IT_011) and test IDs
FR-017 server-clock countdown	3	PR #76	Review asked it to reuse the attempt services; merge conflict
FR-029 no answer key in Student payloads	2	PR #76	Same pull request as FR-017
NFR-013 CSRF, ORM-only, generic error pages	2	PR #78	Waiting for review
NFR-014 secrets from environment variables	1	Verified, closed by hand	Waiting on #78
NFR-006 review by another member	1	Issue #38	See D-04

Merge conflicts were the main cause. A teammate's editor reformatted whole files on save (`repository/models.py` in PR #71, `attempts/views.py` in PR #75), so every branch touching those files conflicted. The review also found that the student exam page had no control to start an attempt (FR-006 and FR-015 were built by different members); PR #79 added it on 27 Oct.
Corrective action: members turn off format-on-save for this repository (requested in the review of PR #71), and a branch merges `main` into itself before review so conflicts are resolved by its owner.
Affects: the Sprint 1 burndown, which shows the drop on 27 Oct.

D-06 — The Sprint 1 score is computed but not yet shown to the Student

Baseline: the Sprint 1 goal ends with the Student "receiving an automatically computed score".
Built: every submission is evaluated automatically and its score is stored on the attempt (FR-022, FR-023). No screen shows it yet: the Student scorecard and the Faculty results view belong to FR-024 to FR-027, which the Sprint Plan places in Sprint 2. In the Sprint 1 demo the stored score is read from the database.
Affects: Sprint 1 goal; FR-024, FR-027.
Status: open, delivered by Sprint 2 (30 Oct).

D-07 — Test results recorded in pull requests, not in the Test Plan

Baseline: Definition of Done item 3 requires each item's test cases in `Documents/Test_Plan.pdf` to be executed with their *Actual Result* and *Test Result* columns filled in.
What happened: the test cases were automated and run by CI on every pull request, and each pull request lists its results. The Test Plan's *Actual Result* and *Test Result* columns are still empty. PR #74 names its tests `test_ut028_…` rather than by test-case ID alone, and PR #78 has no results table.
Corrective action: QA & Docs (Komal) fills both columns for every Sprint 1 and Sprint 2 test case from the CI runs before the development freeze on 30 Oct.
Affects: all Sprint 1 test cases; NFR-007 (every FR linked to a test case).